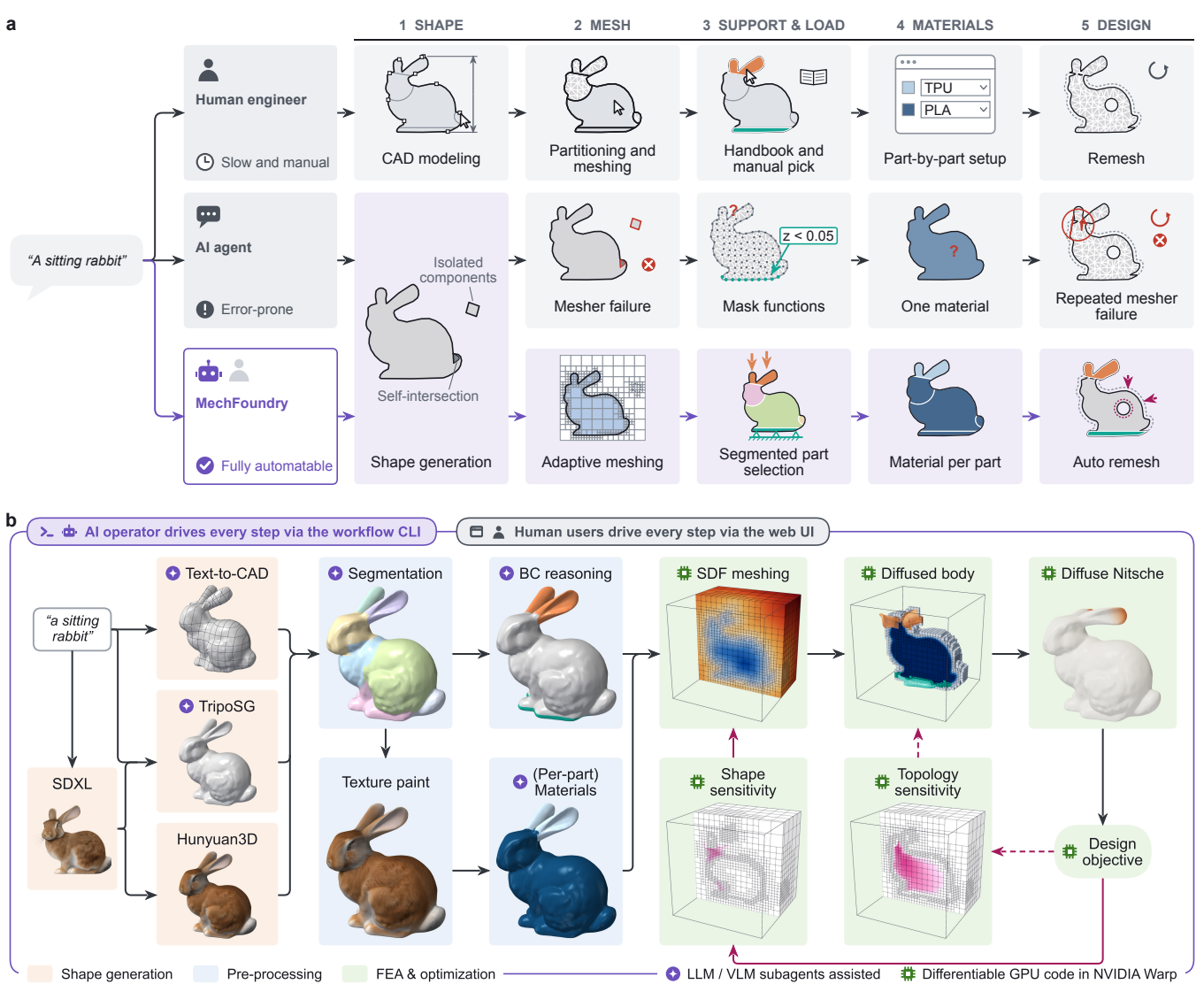

a
1 SHAPE
2 MESH
3 SUPPORT & LOAD
4 MATERIALS
5 DESIGN
"A sitting rabbit"
Human engineer
Slow and manual
CAD modeling
Partitioning and meshing
Handbook and manual pick
TPU
PLA
Part-by-part setup
Remesh
AI agent
Error-prone
Isolated components
Self-intersection
Mesher failure
z < 0.05
Mask functions
One material
Repeated mesher failure
MechFoundry
Fully automatable
Shape generation
Adaptive meshing
Segmented part selection
Material per part
Auto remesh
b
AI operator drives every step via the workflow CLI
Human users drive every step via the web UI
"a sitting rabbit"
SDXL
Text-to-CAD
TripoSG
Hunyuan3D
Segmentation
Texture paint
BC reasoning
(Per-part) Materials
SDF meshing
Shape sensitivity
Diffused body
Topology sensitivity
Diffuse Nitsche
Design objective
Shape generation
Pre-processing
FEA & optimization
LLM / VLM subagents assisted
Differentiable GPU code in NVIDIA Warp